

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester I Examination, 2022-23

Paper No. BPT-101: Mechanics and Relativity

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. (Compulsory — 3.5 marks each for (c)-(f), 3 marks each for (a)-(b). Answer any four parts.)

- (a) Find the amount of work done to increase the speed of an electron from $0.6c$ to $0.8c$.
The rest mass of an electron is 0.5 MeV . [3]
- (b) The volume of a solid does not vary with pressure. Find Poisson's ratio for the solid. [3]
- (c) Explain why a cantilever of uniform cross-section is more likely to break near the fixed end. [3.5]
- (d) Calculate the fictitious force and observed force acting on a body of mass 6 kg kept in frames of reference accelerated vertically upward and downward with acceleration 490 cm/s^2 and 9.8 m/s^2 respectively (take $g = 9.8 \text{ m/s}^2$). [3.5]
- (e) Explain the phenomenon of twin paradox. Is aging reciprocal? [3.5]
- (f) How much faster than its present speed should the Earth rotate so that bodies lying on the equator may fly off into space? [3.5]

Question 2.

- (a) Find the twisting couple per unit twist in the case of a hollow cylinder whose inner radius is R_1 and outer radius is R_2 . [5]
- (b) Derive the relation connecting Young's modulus (Y), Bulk modulus (K) and Poisson's ratio (σ) of a material. [5]
- (c) Calculate the mass of water flowing in 2 minutes through a tube 0.2 cm in inner diameter and 40 cm long, if there is a constant pressure head of 40 cm of water. The coefficient of viscosity of water is 0.0085 CGS units . [4]

Question 3.

- (a) Show that for a compound pendulum the centers of suspension and oscillation can be interchanged. Explain how a compound pendulum is superior to a simple pendulum. [6]

- (b) Derive the equation of motion of a moving coil galvanometer and discuss the conditions under which the motion is (i) deadbeat, (ii) critical and (iii) ballistic. [8]

OR

- (a) Two particles of masses m_1 and m_2 collide and after collision move at angles θ_1 and θ_2 respectively. Show that in the centre of mass frame, the velocities of the particles do not change in magnitude after collision. [8]
- (b) What are the differences between elastic and inelastic collisions? Write down one example for both types of collisions. [6]

Question 4.

- (a) In the experiment of meson decay, how are the observations in the Lab frame and the particle frame explained? Take the half-life of mesons = $2.2 \mu\text{s}$ and velocity = $0.99c$. [7]
- (b) Let the velocity of a particle be v . Discuss the energy distribution in three conditions: (i) $v \ll c$, (ii) $v \sim c$ and (iii) $v \approx c$. [7]

OR

- (a) Consider the relativistic form of Newton's second law of motion. Show that when the force $\vec{F}$ is parallel to the velocity $\vec{v}$:

$$F = m_0 \left(1 - \frac{v^2}{c^2} \right)^{-3/2} \frac{dv}{dt}$$

where m_0 is the rest mass of the object and v is its speed. [10]

- (b) Two spaceships approach each other, moving with the same speed as measured by a stationary observer on the Earth. Their relative speed is $0.70c$. Determine the velocity of each spaceship as measured by a stationary observer on the Earth. [4]

Question 5.

- (a) What is the Coriolis force? Under what conditions does it come into play? [4]
- (b) Discuss in general terms the effect of the Coriolis force produced as a result of the Earth's rotation. [6]
- (c) Calculate the values of the centrifugal and the Coriolis forces on a mass of 20 g placed at a distance of 10 cm from the axis of a rotating frame of reference, if the angular speed of rotation of the frame is 10 rad/s. [4]

[Click here to visit our website](#)
[Click here to visit our contributor](#)
[Click here to visit our contributor](#)